

Team HomeSick

RoomLagbe

Where

Select Radius ▾

Uni Location

Choose Location

Price Range

Min – Max ▾

Search

CSE- 4508
RDBMS PROJECT
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OUR TEAM

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BRIEF OVERVIEW

RoomLagbe is a student focused housing platform designed to help university students find affordable rentals and trustworthy roommates near campus. The system supports:

**Apartment
listings by owners**

**Secure visit request
and approval
workflow**



**Smart search with
filters and commute
time estimation**

**Student-to-student
room sharing**

**Role-based access
for students and
owners**





FEATURES

**User authentication
and role-based
access**

**Apartment and
room-share listing
creation**

**Advanced search
with multiple
filters**

**Commute time
calculation**

**Visit request,
approval, and
rejection system**

**Student and owner
dashboards**

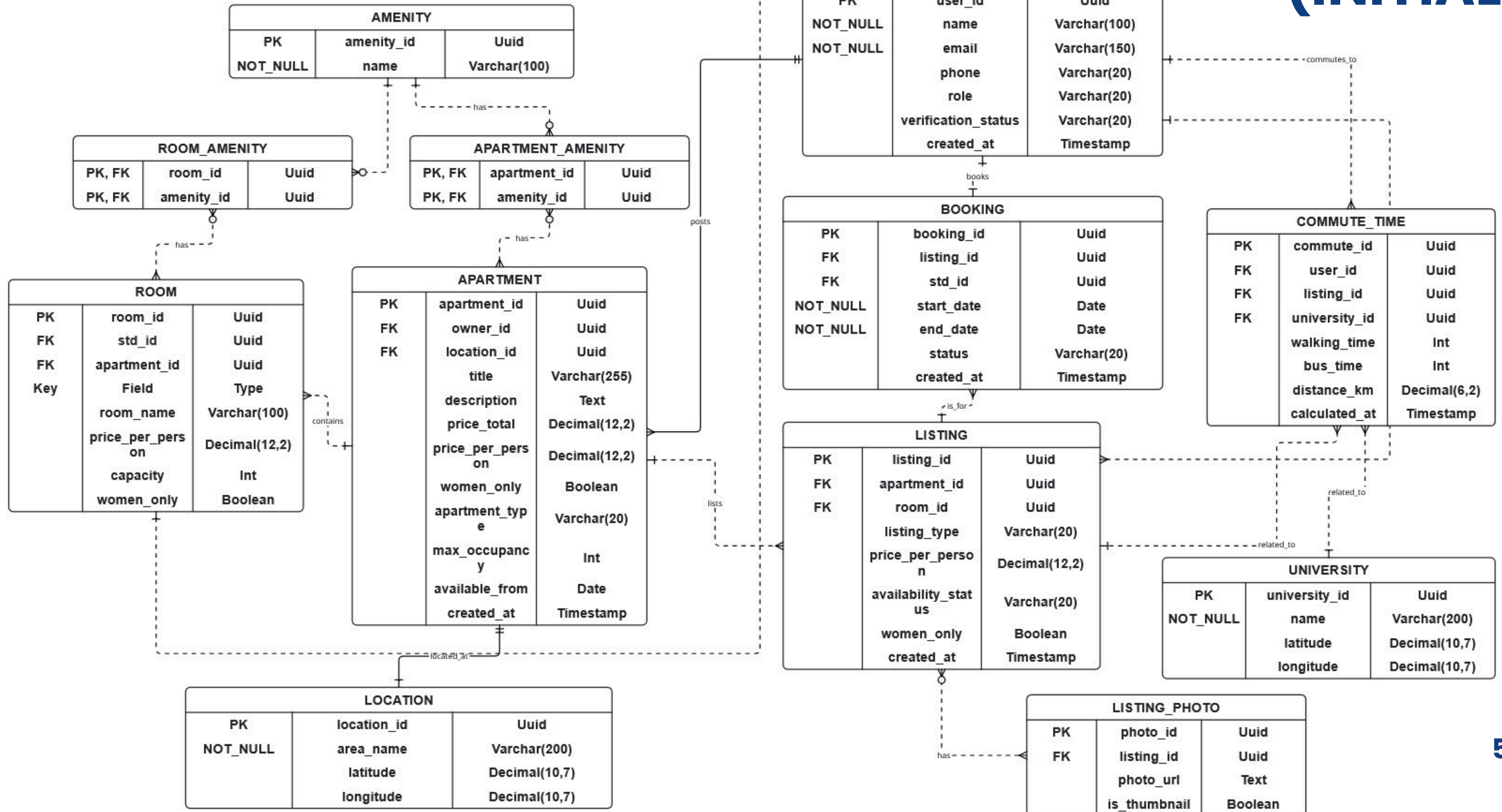
**Wishlist and saved
search functionality**

**Full detailed
listing view**



DATABASE DESIGN

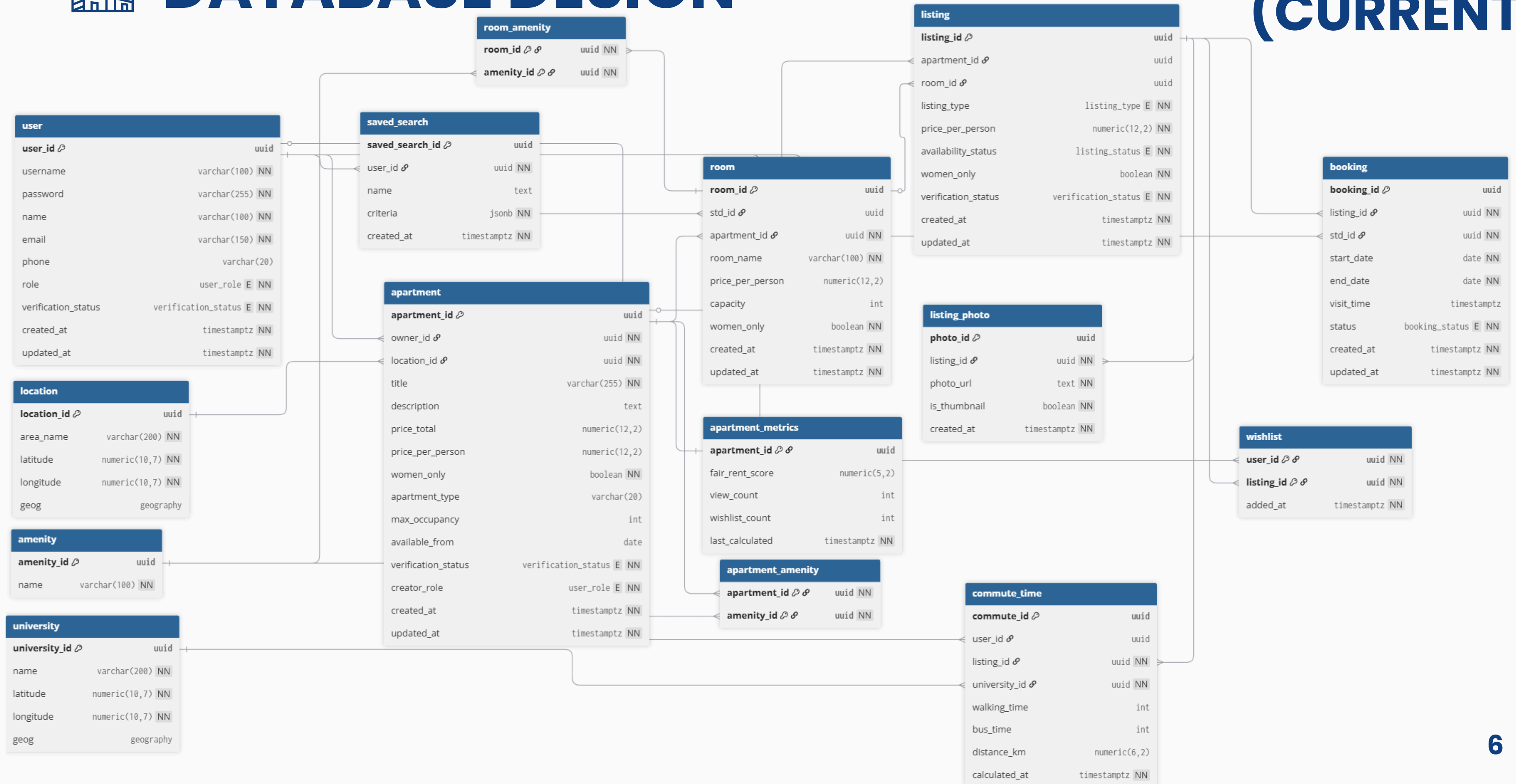
(INITIAL)





DATABASE DESIGN

(CURRENT)





IMPLEMENTED TRIGGERS

Trigger Name	Timing	Table	Purpose
trg_user_update_at	Before Update	User	Automatically updates timestamp whenever a record is modified
trg_location_set_geog	Before Insert or Update	Location	Automatically converts latitude/longitude coordinates into PostGIS geography points.
trg_university_set_geog	Before Insert or Update	University	Converts university coordinates into PostGIS geography points for spatial calculations.
trg_apartment_updated_at	Before Update	Apartment	Tracks when apartment details are modified (price, description, amenities, etc.)



IMPLEMENTED TRIGGERS

Trigger Name	Timing	Table	Purpose
trg_room_updated_at	Before Update	Room	Tracks when room details are modified (price, capacity, etc.)
trg_listing_updated_at	Before Update	Listing	Tracks when listings are modified (status, price, availability)
trg_booking_updated_at	Before Update	Booking	Tracks when booking status changes (pending → approved/rejected)



IMPLEMENTED FUNCTIONS

Type : Trigger

**update_modified_
column**

- Purpose: Auto-updates `updated_at`
- Used in: user, apartment, room, listing, booking

location_set_geog

- Purpose: Convert latitude/longitude → PostGIS geography
- Used in: location

**university_set_
geog**

- Purpose: Convert university coordinates → geography
- Used in: university



IMPLEMENTED FUNCTIONS

Type : Query

get_listings_near_location

- **Type:** Set-Returning Function
- **Returns:** TABLE(listing_id, distance_meters)
- **Purpose:** Radius-based listing search
- **Uses** PostGIS spatial queries

calculate_listing_university_distance

- **Type:** Scalar Function
- **Returns:** Distance in kilometers
- **Purpose:** Show distance between listing & university



POSTGIS SPATIAL FEATURES

PostGIS is an extension for PostgreSQL that adds spatial (geographic) capabilities to the database.

- **Geography columns on location and university tables**
- **Automatic coordinate conversion**
- **Spatial indexes (GIST) for performance**
- **Distance calculations in meters/kilometers**
- **Radius-based search support**



INDEXES & CONSTRAINTS

B-Tree Indexes

- idx_user_username | idx_user_email | idx_user_role
- idx_listing_type | idx_listing_status | idx_listing_price | idx_listing_women_only
- idx_booking_listing_id | idx_booking_user_id
- idx_apartment_owner_id | idx_room_apartment_id
- idx_listing_apartment_id | idx_listing_room_id
- idx_apartment_location_id | idx_commute_listing_id | idx_commute_university_id

Unique Indexes / Constraints

- uq_user_username | uq_user_email | uq_user_phone
- uq_amenity_name | uq_listing_thumbnail | uq_commute_listing_university

Spatial (GIST) Indexes

- idx_location_geog
- idx_university_geog



CURRENT STATUS

COMPLETED

- User authentication and profile management
- Student and owner dashboards
- Apartment and room-share listing creation
- Advanced search and filtering
- Full listing detail view
- Visit booking system with approval/rejection
- 2-hour visit time blocking system
- Wishlist and saved searches

PENDING

- User and owner verification
- Admin workflow and dashboard
- Fair rent analytics
- Map-based apartment/room discovery
- Notifications for students and owners
- Image upload for listings
- Commute time calculation

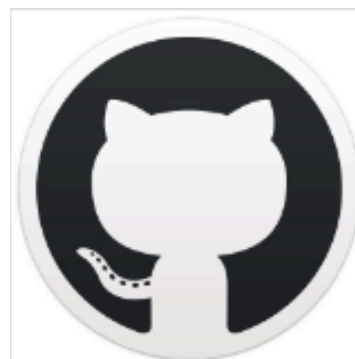


TOOLS & TECHNOLOGY

- **Frontend:** React.js
- **Backend:** Node.js + Express.js
- **Database:** PostgreSQL + PostGIS
(Using Supabase)
- **Authentication:** JWT
- **Styling:** Native CSS
- **Mapping / Commute:** OpenStreetMap API
- **Testing:** Postman
- **Version Control:** Git + GitHub



GITHUB LINK & COMMITS

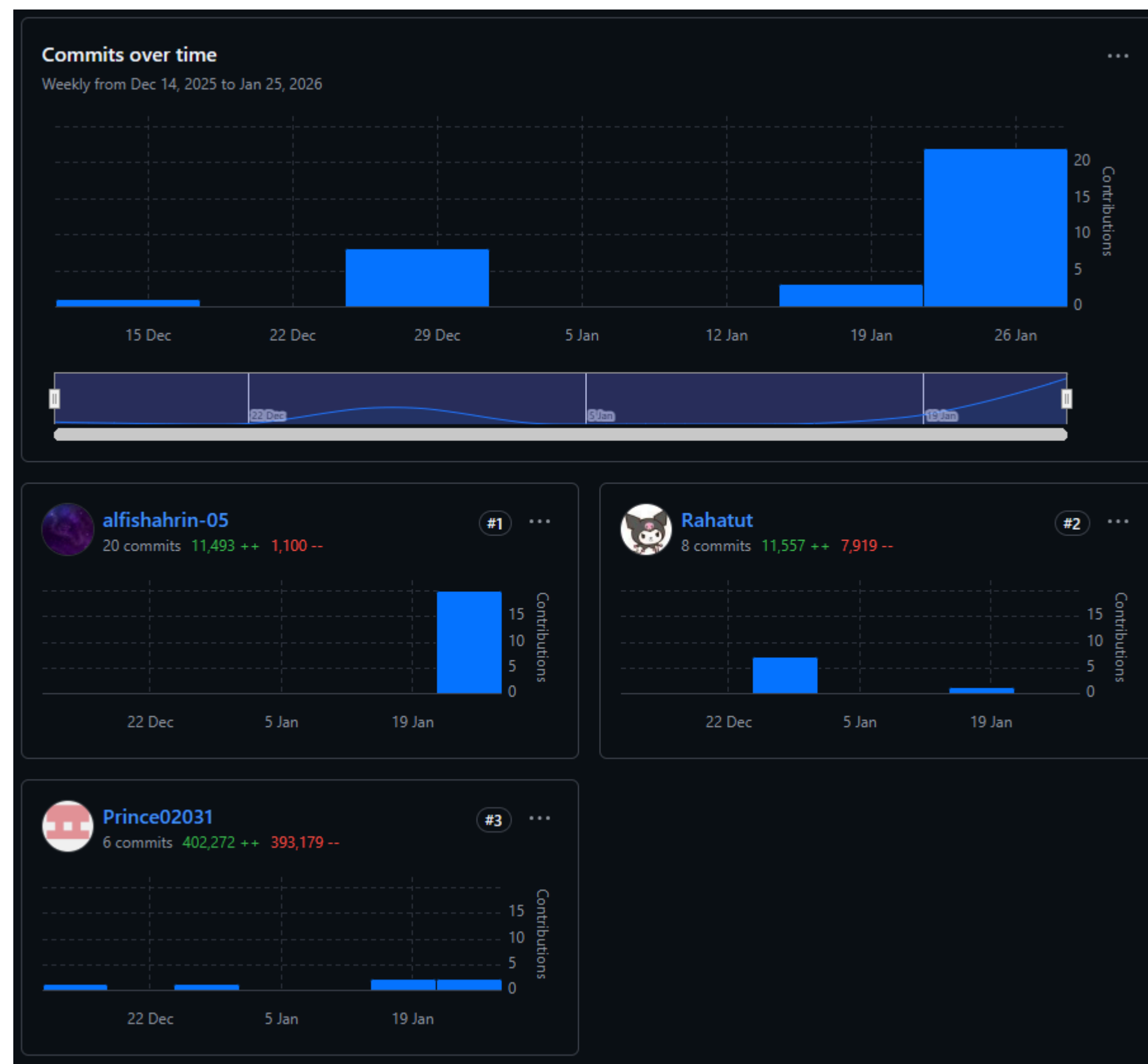


Prince02031/RoomLagbe: RoomLagbe is a student-focused apartment rental platform that enables...

RoomLagbe is a student-focused apartment rental platform that enables landlord listings and student-to-student room...



github.com/Prince02031/RoomLagbe





DEMONSTRATION



INDIVIDUAL CONTRIBUTIONS

Member	Database	Backend	Frontend
Rahatut Tahrim Mounota	Initial Schema design, ER diagram, normalization, indexing	Authentication, Initial Models, Controllers & Routes for core features	Listing creation and display pages (Homepage, apartment/room cards etc)
Nayeemul Hasan Prince	Final Schema design, Triggers, Functions, data integrity & validation rules	Updated Models, Core Business Logic (services) and Backend DB Integration	Owner Dashboard, Listing Details page, Amenities Selection & Location Dropdown
Alfi Shahrin	Functions, Audit Log Design, Views, analytics queries	Bookings & Wishlists, Protected Routes & Postman Collections	Student Dashboard, Booking Interfaces, saved searches, Profile page



THANK YOU